Fengxu Yang

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RESREARCH INTERESTS

My research primarily focuses on **LiDAR-centric perception and the security and privacy of autonomous driving systems**. Previously, I worked on wireless network coexistence, heterogeneous networking, and cross-technology communication to enable robust connectivity for embedded devices.

EDUCATION

ShanghaiTech University Sep. 2022 - July. 2026
Ph.D. in Computer Science and Technology
Supervisor: Zhice Yang

ShanghaiTech University Sep. 2019 - July 2022
M.S. in Information and Communication Engineering
Supervisor: JianMing Wei

North University of China Sep. 2014 - July 2018
BEng. of Information Countermeasure Technology

EXPERIENCE

Shanghai Advanced Research Institute, Chinese Academy of Sciences Feb. 2019 – Sep. 2019
Research Assistant, LoRa Testbed Project

PUBLICATIONS

Conference:

1. **(RAID'25, CCF-B, Core A)** **Fengxu Yang**, Paizhuo Chen, Yihui Yan, Zhice Yang, "An In-model Spy in Edge Intelligence", The 28th International Symposium on Research in Attacks, Intrusions and Defenses, 2025
2. **(Sensys'25, CCF-B, Core A*)** **Fengxu Yang**, Zaizhou, Jiaqi Zhou, Zhice Yang. "LiDARMarker: Machine-friendly Road Markers for Smart Driving Systems", The 23rd ACM Conference on Embedded Networked Sensor Systems, 2025
3. **(INFOCOM'25, CCF-A, Core A*)** Si Liao, **Fengxu Yang**, Huangxun Chen, Zhice Yang. "Enable Autonomous Backscatter in Everyday Devices", IEEE International Conference on Computer Communications, 2025
4. **(IPSN'22, CCF-B, Core A*)** **Fengxu Yang**, Pei Tian, Xiaoyuan Ma, Carlo Alberto Boano, Ye Liu, Jianming Wei. "EMU: Increasing the Performance and Applicability of LoRa through Chirp Emulation, Snipping, and Multiplexing", The 21st ACM/IEEE International Conference on Information Processing in Sensor Networks, 2022
5. **(EWSN'21, Core A)** Pei Tian, Xiaoyuan Ma, Carlo Alberto Boano, Ye Liu, **Fengxu Yang**, Xin Tian, Dan Li, Jianming Wei, "ChirpBox: An Infrastructure-Less LoRa Testbed", The International Conference on Embedded Wireless Systems and Networks, 2021

Journal:

1. **(Sensors'21)** Qingjie Guo, **Fengxu Yang**, Jianming Wei, "Experimental Evaluation of the Packet Reception Performance of LoRa", Sensors, 2021, 21(4): 1071.

Whorkshop & Demo & Poster:

1. **(Sensys'25)** Zaizhou Yang, Yihui Yan, **Fengxu Yang**, Zhice Yang, "Demo Abstract: Using Fingerprint Scanner for On-Body Messaging", The 23rd ACM Conference on Embedded Networked Sensor Systems, 2025
2. **(Sensys'25)** Yihong Hang, Hao Li, Huangxun Chen, **Fengxu Yang**, Zhice Yang, "LLM-Piloted Visual Privacy Agent on Mobile Systems", The 23rd ACM Conference on Embedded Networked Sensor Systems, 2025
3. **(EWSN'22)** **Fengxu Yang**, Pei Tian, Xiaoyuan Ma, Carlo Alberto Boano, Ye Liu, Jianming Wei. "Demo: Real-Time Decoding of LoRa Packets Without Prior Knowledge of their Spreading Factor", The International Conference on Embedded Wireless Systems and Networks, 2022

4. **(DATA'21 (SenSys Workshop))** Pei Tian, **Fengxu Yang**, Xiaoyuan Ma, Carlo Alberto Boano, Xin Tian, Ye Liu, Jianming Wei. "Dataset: Environmental Impact on the Long-Term Connectivity and Link Quality of an Outdoor LoRa Network", The 19th ACM Conference on Embedded Networked Sensor Systems, 2021
5. **(EWSN'21)** Xiaoyuan Ma, Dan Li, **Fengxu Yang**, Carlo Alberto Boano, Pei Tian, Jianming Wei. "Poster: Chirpbox-A Low-Cost LoRa Testbed Solution", The International Conference on Embedded Wireless Systems and Networks, 2020

Patent:

1. **Fengxu Yang**, Yu An, Bobo Zhou, Shaoyu Wang, Huitong Fu, "Elderly Mobility aid Monitoring Device Based on the BeiDou Satellite Navigation System." CN206479642U Granted

HONORS & AWARDS

- ACM SIGBED Student Travel Grant, CPS-IoT Week (10/110), 2025
- SIST Outstanding Teaching Assistant, ShanghaiTech University, 2023
- Second Prize, Future Light – Future Network Technology Innovation Competition, 2022
- EWSN Best Paper Nominee, 2021
- EWSN Best Poster, 2020
- National Undergraduate Innovation and Entrepreneurship Training Program, 2017
- Third Prize, International Underwater Robot Competition, 2017
- First Prize, Shanxi Province Undergraduate Electronic Design Contest, 2016
- Third Prize, National College Student Smart Internet Innovation Competition, 2016
- Third Prize, International Underwater Robot Competition, 2016
- Third Prize, China Robot Competition, 2016
- First Prize, International Underwater Robot Competition, 2015
- Second Prize, National Undergraduate Electronics Design Contest, 2015
- Second Prize, Robot Competition for College Students in Five Provinces of North China, 2015

TEACHING EXPERIENCE

Teaching Assistant

Computer Network (CS120) 2022 Fall, 2023 Fall, 2026 Fall
Wireless and Mobile System (CS222) 2024 Spring
Introduction of Information Science and Technology (SI100B) 2020 Spring

PROFESSIONAL SERVICE

Reviewer

IEEE Transactions on Green Communications and Networking (TGCN), 2026
IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS), Reviewer, 2023